SMOLVLA / LIBERO | 150-EPISODE EXPERIMENT

How often should a robot replan?

Three tasks, ten initial states, five replanning intervals

We vary the number of actions K executed before obtaining a fresh observation and replacing the plan. The study measures task completion, motion, grasping behavior, model calls and runtime for one pretrained model.

OBSERVATION
2 cameras + state + instruction

>

PREDICT
H = 50 actions

>

EXECUTE
First K actions

>

REPLAN
New observation; replace tail

Experimental design

Factor	Setting	Comparison
Tasks	Drawer, soup can, black bowl	Three different manipulation behaviors
Initial states	0-9 for each task	The same ten states for every K
Replanning interval	K = 1, 5, 10, 25, 50	30 episodes per K; 150 in total

All 150 episodes contribute to the main analysis. Each setting has the same task and initial-state coverage; differences reflect the executed prefix length K under the fixed protocol.

Task	Suite	Step limit
Soup can into basket	libero_object	280
Black bowl onto plate	libero_spatial	280
Open the middle drawer	libero_goal	300

Model and controlled conditions

SmolVLA uses flow matching to predict H=50 actions from two images, robot state and a language instruction. Each action specifies relative end-effector translation, rotation and a gripper command. K is an executed prefix length, not repeated execution of one action.

K=1,5,10,25,50; one policy seed (11); 20 Hz simulated control; 10 generation iterations; batch=1; precision fp32; 10 settling steps; 2 warmup calls. Checkpoint, preprocessing, simulator, hardware and logging are held fixed. Physical initial states and initial observations match across K by SHA-256.

Run order is randomized within each source run. Model sampling noise is tied to policy seed and simulation step. The simulator pauses during inference.

Task completion across 150 episodes

3 tasks x 10 initial states x 5 K values. Each K has 30 episodes; each task/K has 10. Success is LIBERO's native signal before the step limit.

K	Drawer	Soup can	Bowl	Total	95% interval
1	9/10	6/10	8/10	23/30	[63.3, 90.0]%
5	10/10	9/10	7/10	26/30	[73.3, 96.7]%
10	10/10	10/10	4/10	24/30	[70.0, 90.0]%
25	1/10	6/10	4/10	11/30	[20.0, 53.3]%
50	3/10	3/10	3/10	9/30	[13.3, 46.7]%

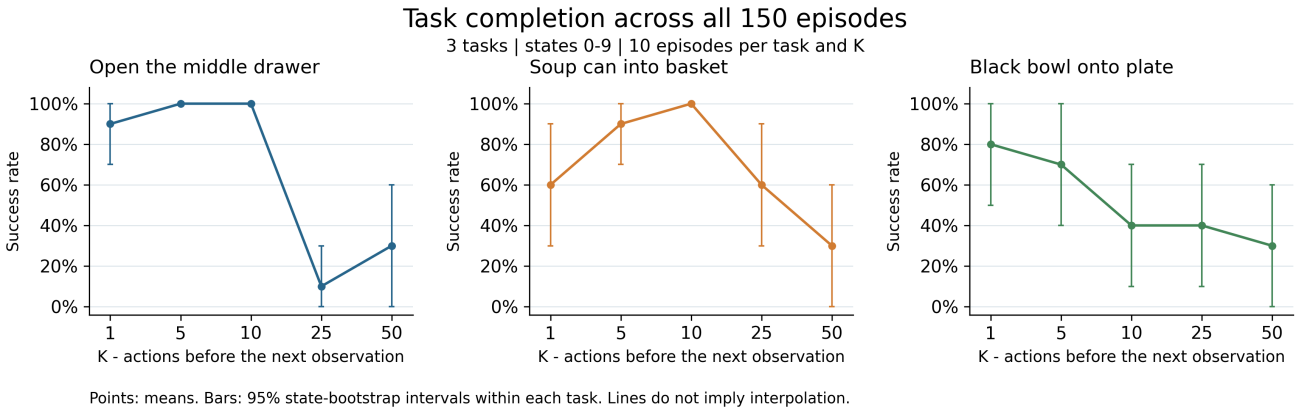


Figure 1. Success by task across all ten initial states. Every K is evaluated from the same states within each task.

Matched contrasts against K=10

K	Success difference, percentage points	95% paired interval
1	-3.3	[-20.0, 13.3]
5	+6.7	[-10.0, 20.0]
10	+0.0	[0.0, 0.0]
25	-43.3	[-60.0, -26.7]
50	-50.0	[-70.0, -30.0]

Means weight tasks equally, then initial states equally. 5,000 bootstrap resamples draw initial-state clusters with replacement within each task; all K outcomes stay paired in every draw. The contrast interval is computed from paired differences, not by subtracting two marginal interval bounds. Tasks and the policy seed remain fixed. Intervals are exploratory and are not adjusted for multiple comparisons.

Which initial situations change outcome?

Each row below holds the task and initial state fixed while K changes. Each cell shows one native LIBERO outcome, not a probability. Reading across a row makes disagreements visible that an aggregate success percentage can hide.

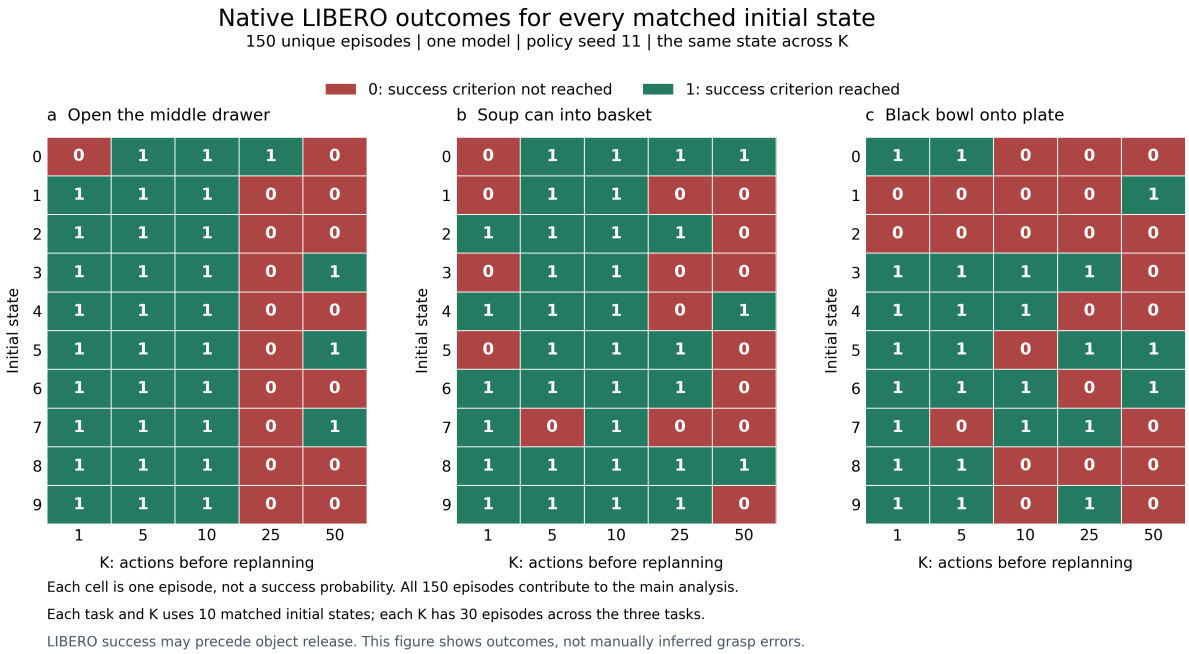


Figure 2. All 150 episode outcomes. Green cells reach the simulator's success condition; red cells reach the step limit without success.

Paired outcomes compared with K=10

K	K succeeds; reference fails	K fails; reference succeeds	Same outcome
1	4	5	21
5	4	2	24
10	0	0	30
25	2	15	13
50	2	17	11

No interval succeeds in every initial situation. The result depends on the task and initial geometry; different settings can have similar totals while failing in different scenes. The paired intervals on page 2 quantify uncertainty across these states, conditional on the fixed tasks and policy seed.

The examples on page 6 examine visible behavior in selected recordings. They illustrate particular failures and recoveries without replacing the complete outcome grid.

Computational cost and robot motion

Runtime values below are means per episode. Early termination affects calls, duration and path length; calls = ceil(executed steps / K). A low cost is not a substitute for successful completion.

K	Calls	Model, s	Simulator, s	Sum, s	Episode, s
1	162.0	125.2	3.6	128.8	132.5
5	31.3	24.3	3.4	27.7	31.7
10	15.5	12.0	3.5	15.5	19.0
25	9.6	7.5	5.0	12.5	16.9
50	5.1	4.1	5.2	9.3	14.0

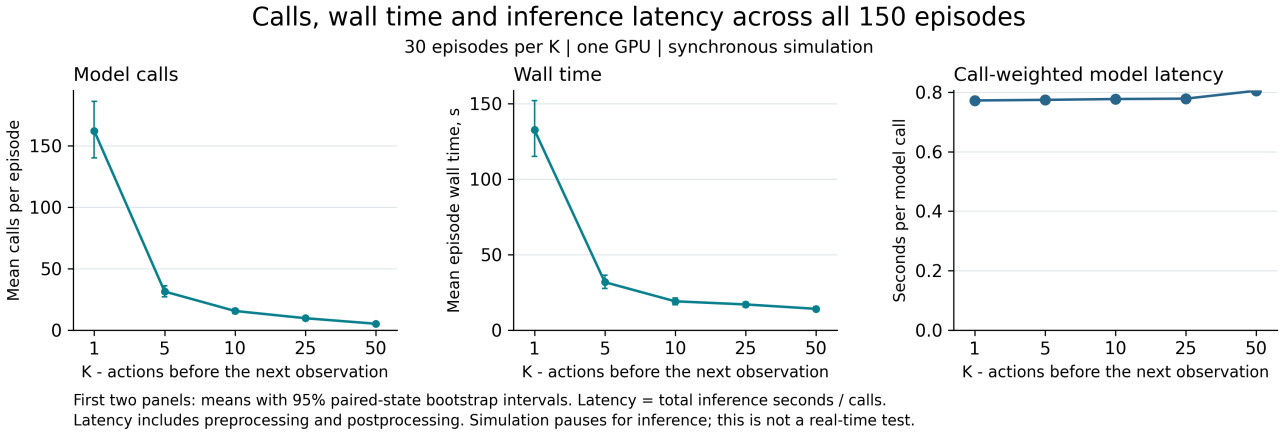


Figure 3. Calls and wall time across all 150 episodes, with state-bootstrap intervals; model latency is total inference time divided by total calls, without an uncertainty interval.

Motion proxies

K	EEF path, m	Plan revision	Replan jump	Gripper closes
1	0.88	0.317	0.337	9.4
5	0.98	0.320	0.334	4.3
10	1.06	0.417	0.440	5.6
25	1.38	0.527	0.534	4.6
50	1.29	undefined	0.455	3.4

Plan revision: L2 difference between the new first translation command and the next command in the previous plan. At K=50 the previous plan is exhausted, so revision is undefined. Replan jump: L2 change between consecutive executed translation commands at replanning. Both use action units, not meters. Changes and gripper-closing transitions are proxies; they do not count grasp errors or prove useful correction. Each proxy mean uses only episodes where that value is defined.

Model time includes preprocessing, inference, postprocessing and GPU synchronization. Simulator time includes stepping and rendering. Episode wall time adds reset, frame recording and video finalization, but excludes model loading, warmup and JSON-log writes. Simulated duration is steps / 20, not wall time.

Measured totals: 6,707 model calls and 107.1 min summed episode time across all 150 episodes. Setup and transfer time are excluded.

Visual annotation: coverage and errors

The review subset contains 60 episodes: every task and K for states 0-3. 60 recordings were inspected; both missed-grasp and drop counts are resolvable in 41. States 4-9 contribute quantitative outcomes and cost but are outside this event-annotation subset. Unresolved contacts remain unknown.

All episodes	Selected	Inspected	Both counts known
150	60	60	41

Review coverage and observed events by K

K	Selected / all	Inspected	Both known	Misses: sum [n/all]	Drops: sum [n/all]
1	12/30	12	7	0 [7/30]	0 [7/30]
5	12/30	12	10	0 [10/30]	0 [10/30]
10	12/30	12	9	0 [9/30]	0 [9/30]
25	12/30	12	8	0 [8/30]	2 [9/30]
50	12/30	12	7	1 [7/30]	0 [7/30]

Inspected counts come only from inspection.json. If no inspection record is supplied, the report says unknown; it does not infer inspection from a CSV row. reviewed=yes denotes both event counts being resolvable. reviewed=no can mean an inspected but ambiguous episode. Each event has its own known denominator: a drop may be clear while missed grasps remain unknown. Unknown events are never zero.

Correction labels in the selected subset

Selected episodes	Yes	No	Unclear	Unavailable
60	11	11	38	0

Review method and interpretation

The assistant reviews both camera views using sampled frame sheets, transitions of the gripper command, the final action and outcome frames; ambiguous contacts receive denser frame checks where available. Exact reviewed steps and evidence are recorded in notes and inspection metadata. There is no independent human review. Counts are observations from this subset, not estimated rates for all 150 episodes.

A missed grasp means closing without retaining the target. Failure to hook a drawer handle with an open gripper is not a missed finger grasp under this definition. A drop requires unintended loss after retention; planned release during placement is excluded. A correction label requires visible redirection plus relevant replanning steps in logs; temporal alignment does not establish a causal benefit.

Illustrative cases: behavior and errors

These paired examples use the same initial situation within each task. Outcomes below come from episode summaries; the accompanying observations are grounded in reviewed videos and logs.

1. A lost can is not recovered

Soup can into basket; state 3; seed 11. K=10: success, 139 steps, 14 calls; K=25: failure, 280 steps, 12 calls.

Observation. K=10 retains the can during transport and reaches native success at step 139. K=25 lifts it, then loses it around steps 82-86 outside the basket. The hand continues toward the basket after replans at 100 and 125; it does not retrieve the can before the step-280 timeout.

Hypothesis. Slippage or contact geometry may explain the loss. A longer interval may leave fewer opportunities to adjust the grasp, but the recordings do not isolate the cause of the drop.

Evidence. Object/state3/seed11; both videos and review-object.json. Dense frames 76-94; gripper commands stay positive (0.987-1.000), so the loss is not a commanded release.

2. A second approach still misses placement

Black bowl onto plate; state 3; seed 11. K=1: success, 80 steps, 80 calls; K=50: failure, 280 steps, 6 calls; K=10: success, 75 steps, 8 calls.

Observation. K=50 withdraws at steps 80-100, returns at 105-125 after replan 100, transports the bowl at 130-160 and opens near the plate rim around 160. Placement remains incomplete at 280. K=1 succeeds at 80 using 80 calls; K=10 also succeeds at 75 using only 8 calls.

Hypothesis. A new plan can redirect an approach without ensuring accurate placement. The K=1 versus K=10 contrast shows that extra updates did not improve this binary outcome; it does not establish general equivalence.

Evidence. Spatial/state3/seed11; videos, steps/chunks and review-bowl.json. Replan 100 command jump 0.742472. Initial K=50 contact counts remain unknown.

3. A longer plan opens the drawer sooner

Open the middle drawer; state 3; seed 11. K=5: success, 252 steps, 51 calls; K=50: success, 127 steps, 3 calls.

Observation. Both episodes succeed. At K=5, the hand remains near the handle around steps 90-215 with little opening, then finishes at 252 after 51 calls. K=50 visibly pulls around 100-127 and succeeds at 127 after 3 calls. Measured compute time is 44.7 versus 4.9 seconds.

Hypothesis. A longer action sequence may sustain an effective pull in this state. Contact forces are not logged, so the delay at K=5 cannot be attributed uniquely to replanning.

Evidence. Goal/state3/seed11; matched initial-state and observation hashes; both videos, summary logs and review-drawer.json. Open-finger hooking is not a missed closing grasp.

Examples illustrate mechanisms and uncertainty; they do not replace the complete outcome table. A REPLAN marker shows plan replacement, not proof that a new image caused an improvement.

Findings and limits of inference

Findings from the complete experiment

1. Across all 150 episodes, K=1, 5, 10, 25 and 50 succeed in 23/30, 26/30, 24/30, 11/30 and 9/30 matched situations. K=5 has the highest observed success count, while its 27.7 s mean compute is 4.7 times lower than the 128.8 s at K=1.

2. Relative to K=10, paired success differences are -3.3 percentage points for K=1 and +6.7 for K=5, with exploratory 95% intervals [-20.0, 13.3] and [-10.0, 20.0]. These intervals include zero: the shorter intervals cannot be reliably ranked or declared equivalent. K=10 uses 15.5 s mean compute.

3. K=25 and K=50 have lower aggregate success. Their paired differences from K=10 are -43.3 and -50.0 percentage points, with exploratory 95% intervals [-60.0, -26.7] and [-70.0, -30.0]. Individual outcomes remain task- and state-dependent, so longer intervals can still succeed where another setting fails.

4. Reviewed videos show an unrecovered can drop, a repeated bowl approach that misses placement, and a successful drawer pull with fewer calls at K=50. Two state-2 drawer failures open the upper drawer instead of the middle one. Visible redirection after replanning does not guarantee recovery; contact-mechanism explanations remain hypotheses.

What the comparison identifies

Changing K changes both the observation interval and how often stochastic action chunks are regenerated. The comparison identifies the effect of the execution policy K; it cannot isolate fresh visual feedback from plan replacement or stochastic regeneration. Matching sampling noise at shared simulation steps does not create common subsequent trajectories.

Remaining limitations

The experiment covers ten states per task, but only three selected tasks, one checkpoint and one policy seed. Bootstrap intervals describe variability across these states conditional on those choices. They do not estimate variation across new tasks, models, seeds or physical robots. The multiple K comparisons are exploratory. These simulator states are not claimed to be unseen during checkpoint training.

Native LIBERO success can occur while an object is still held in the target region. Episodes stop at that signal; stable placement after explicit release was not tested separately. Early termination also changes cost and path length. Visual event counts cover a subset and retain unresolved contacts as unknown.

Physics pauses while inference runs. A 20 Hz simulated controller is not evidence of real-time operation under model latency. Sessions can also differ in system load. No conclusion about a physical robot is supported by these simulations alone.

Conclusion

Replanning frequency changes computational cost and task behavior. K=5 and K=10 offer useful observed cost-success compromises in this setup. Updating after every action increases compute without exceeding the success count at K=5. Longer intervals reduce model calls but lose aggregate success, with exceptions in individual situations. No universal optimum follows from three tasks, ten initial states per task and one policy seed. Native success, assistant-only event annotation and synchronous simulation limit claims about stable placement, causal correction and real-time robots.

Next controlled tests

Multiple policy seeds and additional tasks would test robustness. A separate intervention comparing fresh and stale observations at the same regeneration schedule would help separate feedback from chunk replacement. Adaptive K and physical-robot latency require their own evaluation protocols.

Reproducibility and source provenance

Device: NVIDIA GeForce RTX 2060; CUDA 12.6; Python 3.12.3; Linux. LeRobot 0.6.1; hf-libero 0.1.4; MuJoCo 3.3.2; PyTorch 2.7.1.

Model: HuggingFaceVLA/smolvla_libero
Revision: 6721902bc4d61e50a3bfdb11dfb4cb626f05d102
Backbone: HuggingFaceTB/SmolVLM2-500M-Instruct
Revision: 7b375e1b73b11138ff12fe22c8f2822d8fe03467

Rebuild results from the repository root

```
python scripts/analyze_extension.py
python scripts/build_expanded_figures.py --overwrite
python scripts/build_state_outcomes.py --overwrite
python scripts/build_expanded_report.py --overwrite
python scripts/build_expanded_demo.py --overwrite
```

See docs/linux-checklist.md for collecting the episodes and setting up the pinned runtime.

The analysis requires complete paired episodes and matching runtime, checkpoint and frozen core code. Finish visual review and save annotations.csv plus inspection.json before rebuilding statistics. Then author discussion.json from actual videos. Plot regeneration after an edit uses --overwrite. PDF generation needs ReportLab and pypdf, followed by page rendering and visual inspection.

Frozen core SHA-256: d96cfeded0dc6e1ca630d2fbdc9f23e1cdf51806bfbe14211327200da412902e
Dependency lock SHA-256: 8574b51f0fad0b6ec40c02b3c6192da4606e30a94fd8dc1e18068b461d81212a
Expanded statistics SHA-256: 1d3d77bdb4bb85fc5772bc7031fc601a7e432f43655ee612f4511d9be6f2e576
Study discussion SHA-256: 0d93f0cd2163965ee495ec506c133395b44296729b5729a0c4a3f839981e3d5b

The statistics record every source manifest hash and every raw summary, action log, chunk log and video hash. Original episode files and their fingerprints remain unchanged. PDF provenance binds the statistics, discussion and figures to this report through SHA-256 digests.

References

1. SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics
2. Evaluated SmolVLA LIBERO checkpoint (pinned revision)
3. LeRobot 0.6.1 SmolVLA implementation
4. LeRobot 0.6.1 LIBERO adapter
5. LIBERO environment documentation

References describe the model and software. All experimental numbers come from the actual source episodes; interpretation is supplied in the reviewed discussion files.